



OPEN Automatic detection and prediction of COVID-19 in cough audio signals using coronavirus herd immunity optimizer algorithm

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The global spread of COVID-19, particularly through cough symptoms, necessitates efficient diagnostic tools. COVID-19 patients exhibit unique cough sound patterns distinguishable from other respiratory conditions. This study proposes an advanced framework to detect and predict COVID-19 using deep learning from cough audio signals. Audio data from the COUGHVID dataset undergo preprocessing through fuzzy gray level difference histogram equalization, followed by segmentation with a U-Net model. Key features are extracted via Zernike Moments (ZM) and Gray Level Co-occurrence Matrix (GLCM). The Enhanced Deep Neural Network (EDNN), tuned by the Coronavirus Herd Immunity Optimizer (CHIO), performs final prediction by minimizing error metrics. Comparative simulation results reveal that the proposed EDNN-CHIO model improves MSE by 25.35% and SMAPE by 42.06% over conventional models like PSO, WOA, and LSTM. The proposed approach demonstrates superior error reduction, highlighting its potential for effective COVID-19 detection.

Keywords COVID-19 detection and prediction, Cough audio signals, Coronavirus herd immunity optimizer, Enhanced deep neural network

Abbreviations

GLCM	Gray level co occurrence matrix
MSE	Mean square error
ZM	Zernike moments
EDNN	Enhanced deep neural network
CHIO	Coronavirus herd immunity optimizer
SARS-CoV 2	Severe Acute Respiratory Syndrome CoronaVirus 2
SMAPE	Symmetric mean absolute percentage error
AI	Artificial intelligence
COVID-19	COrona VIRUS Disease-19
MFCC	Mel frequency cepstral coefficient
WHO	World Health Organization
MAE	Mean absolute error
CNN	Convolutional neural network
qRT-PCR	Quantitative RT-PCR
DTs	Decision trees
Ct	Threshold cycle
RMSE	Root mean square error
LBP	Local binary patterns
EMD	Empirical mode decomposition
YAMNet	Yet another mobile network
ANN	Artificial neural network
MAPE	Mean absolute percentage error
FD	Fractal dimensions
HIP	Herd immunity population

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ViT Vision transformer
2D Two-dimensional

The recently identified SARS-CoV 2 defines the viral illness known as COVID-19¹. The SARS and the ordinary cold are similar to COVID-19. The WHO states that the moderate COVID-19 signs might involve cough, a fever, and breathing difficulties similar to a cold². Similar to more extreme situations of SARS, COVID-19 also induces pneumonia and/or serious respiratory problems. In extremely rare circumstances, the illness could be deadly; the global fatality rate is thought to be 0.28%³. On December 31, 2019, the earliest occurrences of COVID-19 were originally identified as pneumonia but were subsequently reidentified as COVID-19⁴.

The virus has proved to be extremely contagious, rapidly spreading when in proximity to an affected individual (mean rate of infection of 2.5)⁵. More precisely, it comes from an affected person (created by sneezing and coughing)⁶, as stated by the WHO. After 3 months of the perspectives being recorded in China, on March 11, 2020, the WHO proclaimed it a worldwide outbreak⁷. The health frameworks across the globe are under a great deal of strain because to COVID-19, and even affluent nations struggle to screen sufficient individuals to successfully prevent its growth⁸. Therefore, it is crucial to reconsider the traditional methodologies for rapid case detection and to make the best use of the restricted resources at hand while considering the COVID-19 outbreak scenario into account⁹.

Serology is currently being used for COVID-19 testing, and the virus' genetic information is used to provide a prognosis¹⁰. Clinical molecular diagnostics offer strong diagnostic accuracy, but they also need specialized tools and qualified individuals to be performed¹¹. These procedures can take many days or a few hours to complete¹². Finding the COVID-19 positive patients is of utmost significance in the fight as opposed to the disease's development as the COVID-19 outbreak wreaks havoc on the globe¹³. Assessing both financial and human resource capability is still difficult in several nations¹⁴. Several machine learning algorithms have been created in this regard for the early detection of situations using more expedient and affordable procedures¹⁵.

In spite of the progress in utilizing cough sounds for COVID-19 detection, several gaps persist. Existing approaches often rely on traditional machine learning models that may lack robustness in distinguishing COVID-19 cough sounds from other respiratory conditions due to limited feature extraction capabilities. Most methods focus on single-layered neural networks or basic optimization techniques, which may not achieve optimal prediction accuracy or error minimization. Furthermore, while some studies employ segmentation models, the integration of advanced preprocessing and feature extraction techniques remains underexplored, limiting the overall precision and reliability of COVID-19 detection frameworks. Additionally, many current algorithms are not tuned for enhanced error metrics, impacting their suitability for practical deployment in large-scale screening.

This research aims to develop a comprehensive framework for automatic COVID-19 detection from cough audio signals, overcoming the limitations of prior approaches by integrating sophisticated deep learning and optimization techniques. The proposed framework includes advanced preprocessing and segmentation methods, precise feature extraction, and an optimized prediction model designed for enhanced accuracy and reduced error. The graphical abstract is pictorially represented in Fig. 1.

The paper contribution is.

- To introduce a novel framework for effectively detecting and predicting COVID-19 utilizing deep learning algorithms from cough audio signals.

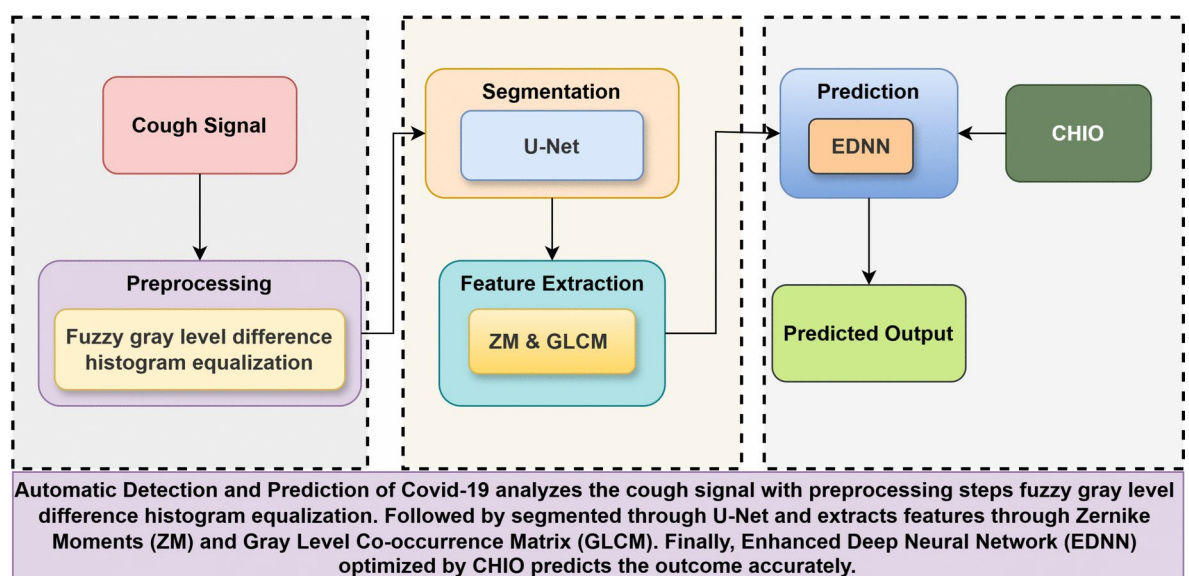


Fig. 1. Graphical abstract.

- To perform the final prediction phase by the EDNN model, where the hidden neurons of DNN are tuned by CHIO with the consideration of error minimization as the objective function.
- To demonstrate the feasibility of the proposed model by comparing it with conventional model with respect to various error metrics.

The novelty of this work lies in the unique combination of an Enhanced Deep Neural Network (EDNN) and the Coronavirus Herd Immunity Optimizer (CHIO). The EDNN's architecture, optimized by CHIO, not only enhances model accuracy but also minimizes error metrics across multiple benchmarks. Additionally, the inclusion of Zernike Moments (ZM) and Gray Level Co-occurrence Matrix (GLCM) for feature extraction represents a novel approach in COVID-19 detection from audio signals, providing greater discriminative power than conventional methods. Comparative results demonstrate the superiority of this framework, highlighting its potential for real-time and scalable COVID-19 screening.

The paper organization is described as follows. Section “[Introduction](#)” is the elaboration of COVID-19 cough model. Section “[Literature survey](#)” is a literature survey. Section “[Proposed model and pre processing for the developed automatic detection and prediction of COVID-19 model](#)” is preprocessing for the developed prediction of COVID-19 model. Section “[Segmentation and feature extraction for the developed automatic detection and prediction of COVID-19 model](#)” is segmentation and feature extraction for the COVID-19 model. Section “[Prediction and optimization for the developed automatic detection and prediction of COVID-19 model](#)” is prediction and optimization for the COVID-19 model. Section “[Results](#)” is results. Section “[Conclusion](#)” is conclusion.

Literature survey

In 2020, Laguarda et al.¹⁶ have created a structure for artificial intelligence language processing for translational patient monitoring in non-invasively, real-time, and at zero variable cost. To produce a binary pre-screening diagnosis, cough samples were processed with three parallel ResNet50 layers and one Poisson biomarker layer. We used the additional 1064 patients in the dataset to evaluate the CNN-oriented models after training them on the 4256 participants in the dataset. The technology's COVID-19 discrimination accuracy was increased by applying transfer learning to acquire biomarker features on bigger datasets that had already been effectively validated in the lab for Alzheimer's.

In 2022, Husain et al.¹⁷ have looked into how AI system uses breathing, cough, and voice records, as stated in the literature, to identify COVID-19 sickness. Here, we list and discuss the characteristics of the discovered AI methods and the datasets that were utilized in their application. According to PRISMA-ScR criteria, comprehensive exploratory examination was conducted. During 1 April 2020 to 15 August 2021, electronic databases (Science Direct, Google Scholar, and IEEE Xplore) were scanned. Words were chosen on the basis of the intended therapy (AI), the intended disease (COVID-19), and the disease's acoustical indicators (i.e., breathing, speech, and cough). The gathered information was summed up using a literary strategy. In 2022, Ponomarchuk et al.¹⁸ have suggested utilizing voice recordings generated on consumer electronics as input for a machine learning technique to certainly find COVID-19. The technique allows COVID identification on coughs by combining signal processing as well as noise reduction techniques with a group of well calibrated deep learning networks. Both datasets from publicly available sources and the noisy data gathered by end users during beta analysis demonstrated the application's strong effectiveness.

In 2022, Perez et al.¹⁹ have used 8380 clinically tested samples with the extent of the illness on the basis of the patients' lymphocyte count and quantitative Reverse Transcription Polymerase Chain Reaction (qRT-PCR) Ct. The suggested generic approach was an EMD-based technique for cough sound identification and classification using a stack of audio sonograms and a deep ANN classifier with convolutional layers named “DeepCough.” Investigations have been done on DeepCough2D as well as DeepCough3D, two variants on the basis of the count of tensor dimensions. ‘CoughDetect,’ a multi-platform prototype web-app, has been developed using these techniques.

In 2022, Sharma et al.²⁰ have suggested that doing the initial evaluation of COVID-19 would benefit from a thorough audio texture classification of the COVID-19 sounds. Three separate COVID-19 signal types, namely the breath, cough, and voice signals, are subjected to texture analysis. The LBP as well as Haralick's feature was recovered from the signal's spectrogram in order to investigate the sounds. The five classes were used for the initial time in the textural evaluation of cough as well as breath sounds. Throughout the studies, 5 classifications were created for 71.7% of the cough specimens and 72.2% of the breath specimens. Moreover, 79.7% of speech data were split up into 2 categories. Binary categorization yielded the best accuracy percentage of 98.9%.

In 2022, Sobahi et al.²¹ have created a unique, comprehensible technique, which could also include various noises. This study considers the deep YAMNet paradigm. To rebuild the basic cough sounds after labeling, the parts marked as cough were clipped and combined. Next, a graphic explanation of the internal dynamics of the ViT concept was provided. In this work, three publicly accessible cough sound datasets—COUGHVID, VIRUFY, and COSWARA—were employed. For the datasets VIRUFY, COUGHVID, and COSWARA, we attained accuracy of 98.45%, 98.15%, and 97.59%, correspondingly. When contrasted to innovative methodologies, the generated system outperformed them all and was now ready to be put to the trial in practical uses. In 2022, Ren et al.²² examined acoustic aspects. Moreover, we use machine learning techniques to train a network for automatic COVID-19 identification and investigate latent features by assessing how each feature affects forecasts of COVID-19 condition.

In 2023, Manzella et al.²³ have considered a collection of samples of volunteer individuals' coughs as well as breaths that are identified with their COVID-19 condition and were initially gathered by the University of Cambridge. We investigate the issue of these recordings' automatic categorization utilizing interval temporal DTs and forests by understanding them as multivariate time series. Here, we implement a symbolic technique

and demonstrate that it outperforms the existing attained with the identical dataset as well as the majority of non-symbolic methodologies implemented on remaining datasets. In addition, we are capable of obtaining relevant information to aid clinicians in characterizing normal COVID-positive cough as well as breath due to the significance of the technique.

In 2023, Mohammad Hassan Tayarani Najaran²⁴ has suggested a method to detect the COVID-19 illness from cough signals. The suggested methodology was an ensemble system made up of many base learners, each of which employs a distinct feature extractor technique, such as statistical techniques or CNN for automated feature extraction. The raw signal's features were tuned according to an evolutionary perspective. This study also suggests a memetic method that integrates evolutionary methods' broad search area coverage with gradient descent's speed for training CNNs in base-learners. A variety of CNN designs from the literature as well as general methods for machine learning were evaluated against the suggested technique in experimentation. The findings showed that the developed technique performs more effectively than current techniques. Cough signals might be processed using CNNs, and CNN design modification might help even more.

In 2023, Altshuler et al.²⁵ have proposed that observed changes of quantitatively recorded cough might forecast clinical outcome and aid in quickly identifying individuals who need surgery or who are likely to die. 123 COVID-19-infected individuals were treated. Up until inpatient rehabilitation, hospitalization, including death, patients' coughs were continually monitored technologically in addition to the clinical manifestations of their illnesses. The health findings were predicted using logistic methods fitted on cough time patterns after the evolutionary biology of cough in hospitalised COVID-19 illness was reported. The below Table 1 describes the features and challenges of the considered state-of-the-art methods.

In 2024, Chadaga et al.²⁶ have used various clinical and laboratory parameters to develop machine plus deep learning algorithms to predict the early severity of the COVID-19 disease. The COVID-19 data was obtained after ethical clearance at two hospitals in Manipal. A number of nature-inspired feature selection techniques have been applied to select the least but most vital markers. Five eXplainable Artificial Intelligence (XAI) techniques have been used to demystify the predictions made by the classifiers. Concerning XAI, c-reactive protein, basophils, lymphocytes, albumin, D-Dimer, and neutrophils were some of the key markers. The models could be adopted in different health facilities to assess the severity of COVID-19 in infected patients beforehand so that timely interventions to alter a poor prognosis were made. The computer aided diagnostic method could also help the healthcare professionals and lessen the pressure on the already strained healthcare system.

In 2023, Chadaga et al.²⁷ have built an AI-based decision support system for mild-moderate COVID-19 diagnosis with a focus on differentiating these cases from other diseases based on demographic and clinical parameters. In this study however, cases of severe COVID-19 infection were not included in the clinical considerations. This was due to the fact that the case fatality rates have markedly improved since the introduction of COVID-19 vaccines. For making a prediction, a custom-built stacked ensemble model, comprised of multiple heterogeneous algorithms, has been employed. Additionally, one-dimensional convolutional neural networks, long short-term memory networks, deep neural networks, and residual multi-layer perceptron, four deep learning algorithms have been tried out and evaluated. Prediction assessments were explained by introducing five explainers such as Shapley Additive Values, Eli5, QLattice, Anchor, and Local Interpretable Model-agnostic Explanations. The highest accuracy reached sixty through thirty-two percent after utilizing Pearson's correlation, and particle swarm optimization feature selection techniques for the final stack. The most pertinent indicators that were beneficial for the confirmation of the COVID-19 infection comprised Eosinophil, Albumin, T. Bilirubin, ALP, ALT, AST, HbA1c, and TWBC. The encouraging findings indicated appropriateness of this decision support system for differential diagnosis of COVID-19 and other related respiratory diseases.

In 2024, Kevser and Kaya²⁸ have specialized in the study of measurable characteristics of cough and have sorted them into affirmative, negative as well as symptomatic types. A novel ensemble learning model based on the Super Learner (SL) was constructed to recognize the disease from cough sound exploiting a variety of

Author	Methodology	Features	Challenges
Laguarta et al. ¹⁶	Transfer learning	It enhances the discrimination accuracy	It does not explore the complementary investigation associated with the coughing characteristics
Husain et al. ¹⁷	AI	It discusses the behavior of the of the discovered AI methods	It does not expand the initial set for the future research
Ponomarchuk et al. ¹⁸	Deep learning networks	It combines signal processing and noise reduction techniques for revealing the strong efficiency of the application	It does not track the progression of the disease
Perez et al. ¹⁹	Deep ANN classifier	It describes an EMD-based technique for identifying and classifying cough sounds	The dataset renders the investigation to be inconclusive
Sharma et al. ²⁰	LBP	It returns the best accuracy percentage	The computed fitness limits from uncertainty
Sobahi et al. ²¹	Deep YAMNet	It can be employed for practical uses	It does not affect the procedure of optimization
Ren et al. ²²	Machine learning	It predicts the output in a precise manner	It does not support vast strategic resource allocation
Manzella et al. ²³	Temporal DTs	It attains relevant information in helping the clinicians for the normal cough categorization	It does not generalize to various datasets
Mohammad Hassan Tayarani Najaran ²⁴	CNN	It speeds up the process of training	
Altshuler et al. ²⁵	Logistic methods	It continuously monitors the patients together with the clinical manifestations of their illness	It does not consider various respiratory diseases

Table 1. Features and challenges of conventional methods.

audio features including Frequency Distribution, Time Domain Features, Spectral Features, Time –Frequency Features. In simple term SL was the stacked Generalization method with k-fold cross-validation installed within it which helps in choosing an optimal learner from the available set and also extended the performance through the combination of the learned models using cross validation. The proposed SL model incorporated algorithms for DT, RF, LR, SVM, ET and k-NN. The numerical outcomes demonstrated that the said model would be applicable to all the other diseases, which might be diagnosed from the sounds made by the respiratory system.

The reviewed literature highlights various attempts to leverage cough audio analysis for COVID-19 detection, highlighting advancements in machine learning and deep learning applications. Studies like those by Laguarta et al. and Husain et al. illustrate the potential of using cough sounds for preliminary COVID-19 diagnosis, while frameworks employing AI and deep learning models have demonstrated promising accuracy in differentiating COVID-19-related coughs from other respiratory conditions. Techniques such as transfer learning, noise reduction, and feature extraction using Convolutional Neural Networks (CNN) and other machine learning methods have been explored to improve diagnostic performance.

However, these approaches encounter significant challenges. Firstly, many of the methods lack advanced feature extraction techniques, limiting the depth of audio signal analysis and reducing their ability to capture subtle distinctions in cough patterns. Additionally, several models struggle with generalizability due to limited dataset diversity, as most studies use a single dataset, which may not cover the full spectrum of demographic or environmental variations. Another critical challenge is the presence of background noise in real-world audio samples, which affects model robustness. Further, while some works utilize optimization techniques, they are often not tailored to minimize specific error metrics comprehensively, which limits predictive accuracy and reliability.

Identified gaps and need for advancement These challenges highlight the need for a more robust framework that integrates sophisticated preprocessing, segmentation, and feature extraction methods. The absence of multi-layered neural networks optimized specifically for minimizing errors across multiple metrics is a noted gap. Thus, there is a need for a more advanced model that not only addresses the precision of feature extraction and classification but also employs an optimization algorithm capable of enhancing model performance in real-world environments.

Proposed model and pre processing for the developed automatic detection and prediction of COVID-19 model

Proposed model

The following three crucial processes are made possible by the audio datasets (speeches), and they all contribute to the identification as well as diagnosis of COVID-19: (1) the sounds of coughing may help determine COVID-19 positive cases. (2) Stress recognition. (3) The speed of breathing. Nevertheless, the aforementioned speech-oriented COVID-19 approaches work need for significant efforts to amass sizable data sets. Hence, telemedicine services or smart phone applications may help these processes. The gathered audio signal is preprocessed using the fuzzy gray level difference histogram equalization technique. These pre-processed signals undergo the segmentation phase using the U-Net model. From these segmented images, the features are extracted utilizing the ZM and GLCM techniques. These extracted features undergo the final prediction phase that is accomplished by the EDNN model, in which the hidden neurons of DNN are tuned by CHIO with the intention of error minimization as the objective function. The proposed automatic detection and prediction of COVID-19 model is shown in Fig. 2.

Dataset description

This study uses cough signals to diagnose COVID-19 using the COUGHVID²⁹ data. There are more than 20,000 cough recordings in the collection, together with information on gender, age, location, and COVID-19 diagnosis. 8,407 among the data records were chosen for training as well as classification after the data underwent pre-processing. Of them, 1,283 cases exhibit symptoms, 658 cases have COVID-19, and 6,466 cases are healthy. Due to the presence of noise, such as background noises, etc., certain data were deleted. Subsequently, just the portion of the signal that contained the cough was utilised; the quiet regions had been eliminated.

Pre processing

The preprocessing for the gathered cough audio signal is done by the fuzzy gray level difference histogram equalization technique. For a region S , the coordinates (q, r) is computed as follows:

$$i_h(q, r) = \sum_{n=1}^M \left(\sum_{o=1}^M H_{fud}(q, r) \delta(j(n, o) - h) \right) \quad (1)$$

Here, $h = \{0, 1, 2, 3, \dots, 255\}$. Here, $j(n, o)$ represents the intensity value of the pixel at the coordinate (n, o) in the area S , h shows the grey level value, and δ denotes the Dirac delta function.

Segmentation and feature extraction for the developed automatic detection and prediction of COVID-19 model

Segmentation

The segmentation for the preprocessed signals is performed by the U-Net model. There are two sections to the U-net network: The initial option defines the contracting route, which makes use of a standard CNN structure. A ReLU activation unit, a max-pooling layer, and two consecutive 3×3 convolutions make up every block linked

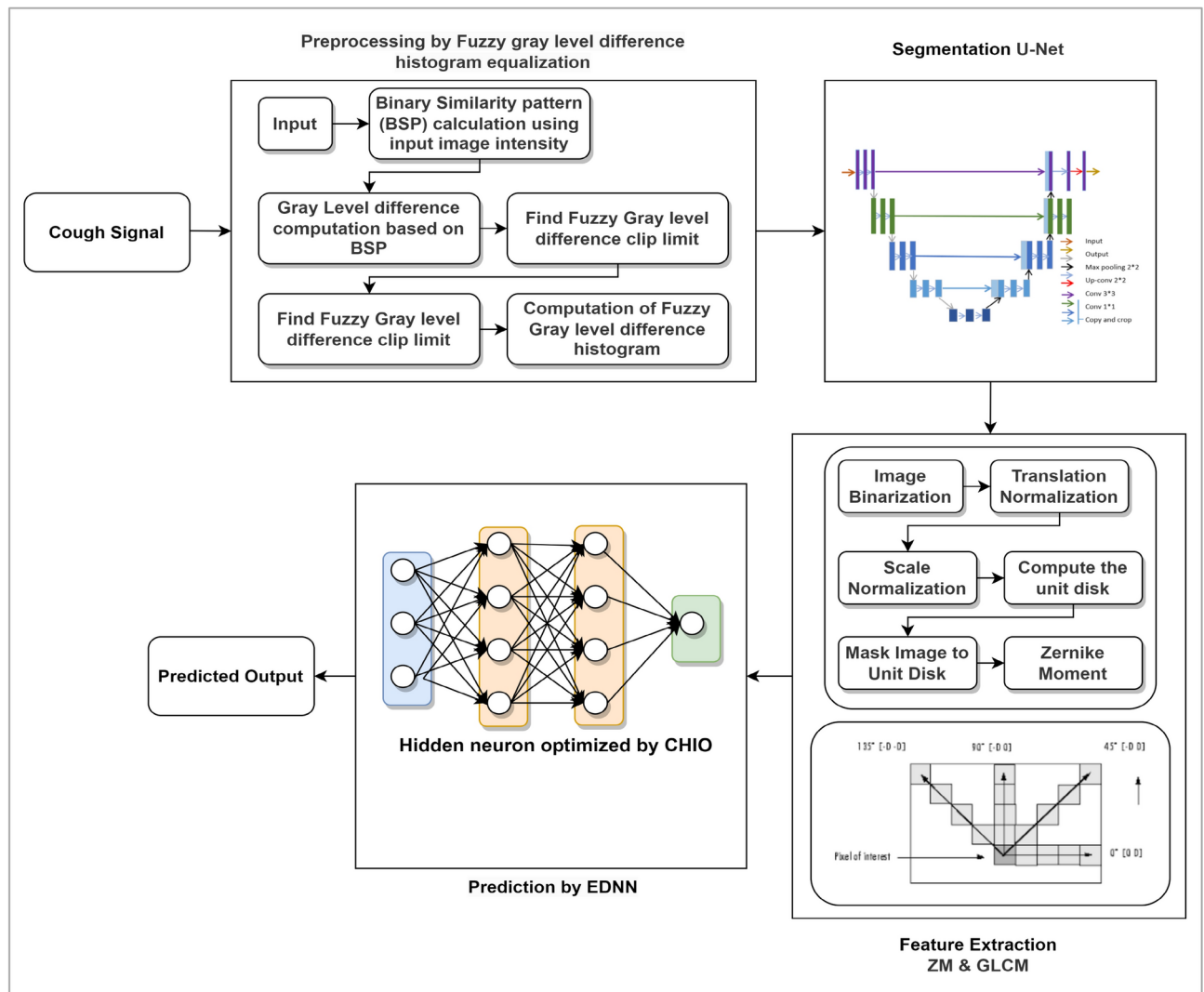


Fig. 2. Proposed COVID-19 prediction model.

with the contracting route. There are various iterations of this pattern. The innovative aspect of U-net is the expanding route, where every step uses 2×2 up-convolution to upsample the feature map.

Two subsequent 3×3 convolutions including ReLU activation come next. For the feature map to be reduced to the necessary count of channels and to create the segmented image, a further 1×1 convolution is used in the last step. As pixel characteristics on the boundaries contain the least number of relevant data, they must be removed, necessitating the reduction. As a consequence, a system with a u-shape appearance is created. Most significantly, relevant knowledge is propagated along the system, enabling it to partition items in an area utilizing context gathered from a broader overlapping region. The channel's energy function is provided by:

$$F = \sum_{i=1}^N x(y) \log(q_l(y)) \quad (2)$$

Here, F shows the energy function, y shows the features, x shows the weight, q_l shows the final feature map's pixel-wise SoftMax function, which is specified as:

$$q_l = \exp(b_l(y)) / \sum_{l=1}^L \exp(b_l(y)') \quad (3)$$

Here, b_l stands for channel activation.

Feature extraction

The features are extracted from the segmented images utilizing ZM and GLCM techniques. The description of both these techniques is provided below.

ZM An extractor on the basis of ZM descriptor may be used to get the parameters. The descriptor must have qualities like (a) give reduced redundancy and increased discriminating capability for direct visualization. (b) depiction of course to finer information (c) rotation, scaling, and translation affine.

Each pixel linked with the pre-processed image contains its ZM. Several polynomial orders (o) as well as the accompanying repetition factor (n) were taken into consideration for the function $J(x, y)$ while computing ZMs. Then, by mapping $J(x, y)$ onto the collection of complex Zernike polynomials as provided in Eq. (4), we estimate the ZM.

$$A_{on} = \frac{n+1}{\pi} \sum_x \sum_y J(x, y) Q_{o,n}^*(x, y), x^2 + y^2 \leq 1 \quad (4)$$

Here, n defines an integer so that $0 \leq |n| \leq o$, and $o - |n|$ represents even, $Q_{on}(x, y)$ represents a Zernike polynomial, $J(x, y)$ represents the image of size 5×5 , and o represents a non-negative integer. The orthogonal basis function $Q_{on}(x, y)$ is described as, while the function $Q_{on}^*(x, y)$ represents its complex conjugate.

$$Q_{on}(x, y) = Q_{o,n}(s, \theta) = S_{o,n}(s) e^{kn\theta} \quad (5)$$

Here, with $0 \leq \theta \leq 2\pi$, $k = \sqrt{-1}$, $s = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$. $S_{on}(s)$, a radial polynomial, is described as:

$$S_{on}(s) = \sum_{r=0}^{o-|n|} (-1)^r \frac{(2o+1-r)!}{r! (o-|n|-t)! (o+|n|+1-t)!} s^{o-r} \quad (6)$$

The complex integer used to describe the ZM obtained from Eq. (4) is depicted in Eq. (7).

$$A_{on} = S_{zm} + kJ_{zm} \quad (7)$$

$$|A_{on}| = \sqrt{S_{zm}^2 + J_{zm}^2} \quad (8)$$

The magnitude associated with the ZM $|A_{on}|$, which represents the ZM descriptor for the given o and n , is represented by Eq. (8). Using o ranging from 1 to 10 and matching combinations of o , n and n so that $0 \leq |n| \leq o$, and $o - |n|$ is even for each pixel, a total of 36 $|A_{on}|$ were produced.

We discover that each of the 36 possible $|A_{on}|$ provides distinct data via the observation of Zernike characteristics. Every point is chosen from both areas with 36 ZM for varied o , n to evaluate the Zernike robust feature (zm_i) as well as Zernike sensitive feature (zm_m). Equations (9) and (10) show the distinction of the zm_i and zm_m .

$$zm_i = \{|A_{0,0}|, |A_{2,0}|, |A_{4,0}|, |A_{6,0}|, |A_{8,8}|, |A_{8,8}|, |A_{10,0}|, |A_{10,8}|\} \quad (9)$$

$$zm_m = \{|A_{1,1}|, |A_{2,12}| \text{ to } |A_{5,5}|, |A_{6,2}| \text{ to } |A_{7,7}|, |A_{8,2}|, |A_{8,6}|, |A_{9,1}| \text{ to } |A_{9,9}|, |A_{10,10}|\} \quad (10)$$

We include an additional feature calculated utilizing the GLCM to the binary classification to improve accuracy. When extracting the GGO as well as pulmonary aggregation features from a CT image, the feature is most significant. The connection among the various angles of image pixels makes up GLCM. If the matrix representing the co-occurrence of grey levels in an image is given by the symbol $r = [r(s, t|e, \theta)]$. In this instance, the GLCM is utilised to calculate the characteristics of the s^{th} as well as t^{th} pixel frequencies by the parameters of length ($e = 1$) together with direction ($\theta = 0$). The technique for calculating the feature represents the identical as for ZM to compute the feature for each pixel.

Entropy, contrast, variance, dissimilarity, correlation, energy, as well as homogeneity represent just a few of the textural characteristics that may be computed using GLCM. We analyzed the GLCM variance as well as contrast as given in Eqs. (11) and (12) within entire of these characteristics. These two characteristics are merged to create the GLCM features Hg , which correctly distinguishes the diseased region from the background region in the CT image.

$$contrast = \sum_{s=0}^{O-1} \sum_{t=0}^{O-1} r(s, t) (s - t)^2 \quad (11)$$

$$variance = \sum_{s=0}^{O-1} \sum_{t=0}^{O-1} r(s, t) (s - \mu)^2 \quad (12)$$

$$Hg_j = [variance_j, contrast_j] \quad (13)$$

If j is the count of training images, μ is the mean, and O is the size of the image, GLCM characteristics are referred to by Hg_j (2 features). Next, as shown in Eq. (14), the features calculated by the ZM as well as the features acquired by the GLCM are merged to generate a feature vector of 38 features.

$$Gt_j = \{zm_{i,j}, zm_{m,j}, Hg_j\} \quad (14)$$

The U-Net model in this work was adapted for spectrogram segmentation to isolate relevant regions in cough audio signals. The contracting and expansive paths of U-Net, consisting of multiple convolutional, max-pooling, and up-convolution layers, were optimized with a learning rate of 0.001 and a batch size of 32. ReLU activation functions were used to introduce non-linearity, while a 1×1 convolution at the final layer reduced the feature map to a single channel for segmentation. This design was chosen for its ability to effectively capture both local and global patterns in the spectrograms.

The CHIO algorithm, inspired by herd immunity principles, was initialized with a population size of 50, a maximum iteration count of 100, and a basic reproduction rate of 1.5. Infected individuals were randomly initialized to simulate the spread of a virtual epidemic. The algorithm balances exploration and exploitation by leveraging social distancing and herd immunity thresholds to adjust gene values dynamically. These initialization strategies were chosen to ensure diverse solution exploration while maintaining computational efficiency.

Zernike Moments (ZM) and Gray Level Co-occurrence Matrix (GLCM) were selected for their ability to capture complementary information. ZM provides rotation-invariant features that describe shape and texture, essential for distinguishing subtle patterns in COVID-19 cough spectrograms. On the other hand, GLCM captures statistical texture properties such as contrast and correlation, which are vital for identifying pixel-level relationships in the segmented regions. Together, these techniques offer a robust representation of the audio features.

Other feature extraction methods, such as Mel Frequency Cepstral Coefficients (MFCC) and Wavelet Transforms, were considered. However, these techniques primarily focus on frequency domain characteristics and do not provide the same level of spatial and textural detail as ZM and GLCM. The combination of ZM and GLCM was specifically chosen to maximize discriminative power and minimize feature redundancy, which was validated by the superior performance of the model across all error metrics.

Prediction and optimization for the developed automatic detection and prediction of COVID-19 model

Prediction

The final prediction phase of the developed cough-19 prediction model is performed by the EDNN model. A DNN represents a feed-forward ANN framework with multiple hidden layers among its input as well as output layers that can categorize features or learn representations. The DNN calculates the probabilities $q_k, k \in \{1, \dots, R\}$ over the classes R when the task calls for classifying a feature y within R groups. Typically, a DNN's input features represent a depiction of the ZM and GLCM. Feature frames are joined together in a sliding window to create the DNN input. DNN creates a nonlinear function for I layers that looks like this:

$$h_x(y) = h_I[X_{IHI}(X_{i-1}, \dots, h_1(X_1y))] \quad (15)$$

Here, $h_i, i = 1, \dots, I$ stands for the activation functions, $X_i, i = 1, \dots, I$ stands for the DNN weights, and y stands for the input features. A DNN trains its weights by minimizing the cost function for cross-entropy given by

$$Z = - \sum_{k=1}^R r_k \log q_k \quad (16)$$

Here, Z stands for the cross-entropy cost function, q_k stands for the Softmax's output, and r_k stands for the equivalent goal. In order to overcome the drawbacks of DNN, the hidden neurons of DNN are tuned by CHIO with the consideration of error minimization as the objective function. This proposed EDNN predicts the final output. The proposed EDNN-based COVID-19 cough prediction model is shown in Fig. 3 below.

CHIO

In order to combat the COVID-19, the herd immunity theory served as the foundation for CHIO. The frequency of immediate communication between sick persons and various members of society determines how quickly the coronavirus illness spreads. Healthcare providers advise social isolation as a means of preventing the sickness from spreading to various people in the community. When the majority of the people are protected, a population is said to have reached a condition of protective immunity, which prevents the spread of illness. These ideas are represented with respect to optimization ideas. Both the herd immunity tactic and the ideas of social estrangement are imitated by CHIO. For herd immunity, three different sorts of individual instances are used: susceptible, infected, and immuned. This will reveal how the recently developed system modifies its DNA via social exclusionary techniques.

The suggested optimization technique models the herd immunity approach. The COVID-19 principles are translated into an optimization framework, and the CHIO is expressed as a series of phases that are in-depth covered here. The following six major phases of the method are covered:

Step 1 Set up the CHIO and optimization issue's parameters. This stage formulates the optimization issue in terms of the objective function as below.

$$\text{Objective function} = \min_y g(y) \quad y \in [LB, UB] \quad (17)$$

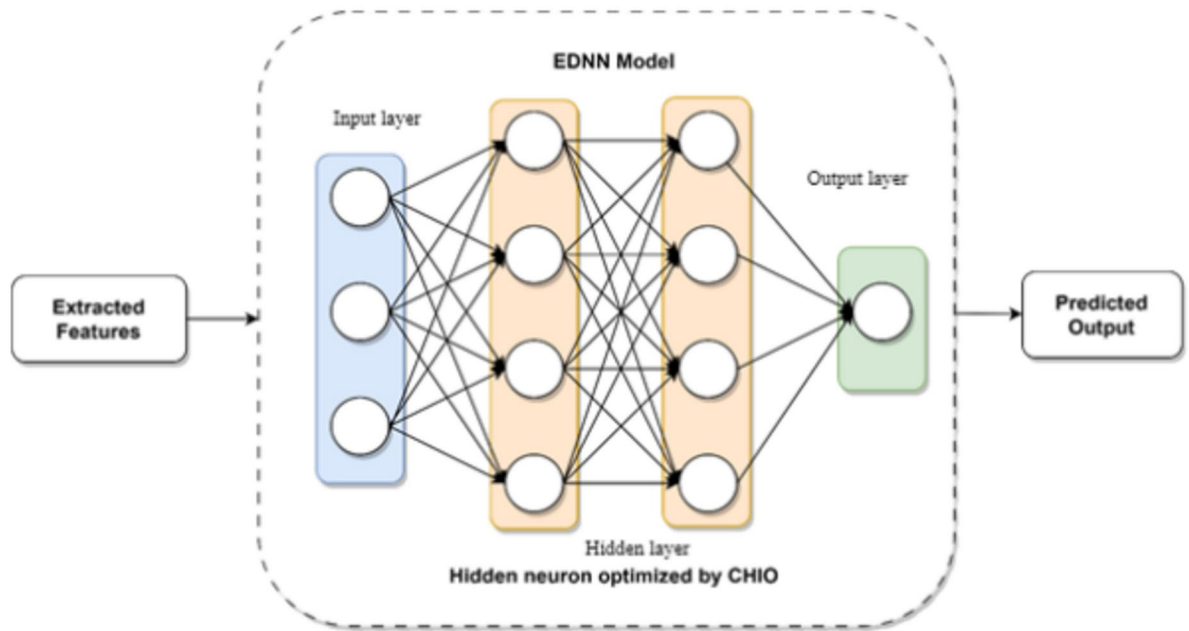


Fig. 3. EDNN-based COVID-19 cough prediction model.

Here, y_j shows the gene (or the choice variable) coded by j and o shows the total count of genes in every person, and $g(y)$ represents the objective function (or immunity rate) derived for the instance (or the person) $y = (y_1, y_2, \dots, y_o)$. Keep in mind that every gene's value range is $y_j \in [LB_j, UB_j]$, in which LB_j as well as UB_j stand for the y_j gene's lower as well as upper limits. CHIO features two control settings and four computational variables. The following four algorithmic variables:

- D_0 : which, in this instance, is started with one, denotes the count of first affected instances.
- $\max_itr$: It represents the most iterations possible.
- IJT : It represents the population.
- o : This represents the dimension of the issue.

In this stage, the CHIO's two primary control parameters must be initialized:

- Basic reproduction rate (CS_s), which regulates CHIO operators by disseminating the epidemic virus among people.
- The phrase “maximum infected cases age” ($\max_{age}$) describes the threshold at which an affected patient either recovers or dies.

Step 2 Create a community with herd immunity. At first, CHIO produces a collection of examples (persons) equal to IJT at randomness (or heuristically). The produced instances are kept in the Herd Immunity Population (HIP) as a 2D matrix of size $o \times IJT$ as below:

$$HIP = \begin{bmatrix} y_1^1 & y_2^1 & \dots & y_o^1 \\ y_1^2 & y_2^2 & \dots & y_o^2 \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{IJT} & y_2^{IJT} & \dots & y_o^{IJT} \end{bmatrix} \quad (18)$$

When every row k stands for an instance y^k , represented by the formula: $y_j^k = LB_j + (UB_j - LB_j) \times V(0, 1), \forall j = 1, 2, \dots, o$. Equation (1) is used to determine the goal function (or immunity rate) for every case. Further, zero (vulnerable case) or one (entire cases) commence the status vector (T) of length IJT for complete cases in HIP (affected case). Keep in mind that (T) has as numerous randomly launched ones as D_0 .

Step 3 Development of coronavirus herd immunity. The primary CHIO loop for development is this one. As per three criteria based on the proportion of CS_s , the gene (y_j^k) of instance y^k either stays identical or it is impacted by social distance as below:

$$y_j^k(u+1) \leftarrow \begin{cases} y_j^k(u) & s \geq BS_s \\ D(y_j^k(u)) & s < \frac{1}{3} \times BS_s // \text{infected case} \\ O(y_j^k(u)) & s < \frac{2}{3} \times BS_s // \text{susceptible case} \\ S(y_j^k(u)) & s < BS_s // \text{immuned case} \end{cases} \quad (19)$$

Here, s produces a count at random among 0 and 1. The following are three rules that may be addressed:

Infected case It is like

$$y_j^k(u+1) = D(y_j^k(u)) \quad (20)$$

$$D(y_j^k(u)) = y_j^k(u) + s \times (y_j^k(u) - y_j^d(u)) \quad (21)$$

Keep in mind that the value $y_j^d(u)$ is picked at random from any affected case y^d depending on the status vector (T) , so that $d = \{j | T_j = 1\}$.

Susceptible case It is shown below.

$$y_j^k(u+1) = O(y_j^k(u)) \quad (22)$$

$$O(y_j^k(u)) = y_j^k(u) + s \times (y_j^k(u) - y_j^n(u)) \quad (23)$$

It is important to keep in mind that the value $y_j^n(u)$ is distributed at random from any vulnerable instance y^n on the basis of the status vector (T) so that $n = \{j | T_j = 0\}$.

Immuned case It is given as follows.

$$y_j^k(u+1) = S(y_j^k(u)) \quad (24)$$

$$S(y_j^k(u)) = y_j^k(u) + s \times (y_j^k(u) - y_j^w(u)) \quad (25)$$

Remember that the value $y_j^w(u)$ is distributed from the optimal immune case y^w on the basis of the status vector (T) , so that.

$$g(y^w) = \arg \min_{k \sim \{l | T_l = 2\}} g(y^k) \quad (26)$$

Step 4 Update the herd immunity level. Every created case $(y^k(u+1))$ immunity rate, $g(y^k(u+1))$, is determined, and the present case, $(y^k(u))$, is substituted by the produced case, $(y^k(u+1))$, if it is better, like $g(y^k(u+1)) < g(y^k(u))$, the age vector B_k is likewise raised by one if $T_k = 1$.

On the basis of the herd immune threshold, which applies the below equation, the status vector (T_k) is modified for every instance y^k :

$$T_k \leftarrow \begin{cases} 1 & g(y^k(u+1)) < \frac{g(y)^k(u+1)}{\Delta g(y)} T_k = 0 \\ 0 & 2g(y^k(u+1)) > \frac{g(y)^k(u+1)}{\Delta g(y)} T_k = 1 \end{cases} \quad (27)$$

Here, $is_corona(y^k(u+1))$ represents a binary value equal to one when the newly created instance $y^k(u+1)$ received a value from any affected case. The $\Delta g(y)$ represents the average immunity rate for a population, like $\frac{\sum_{j=1}^{I \cdot J \cdot T} g(y_j)}{I \cdot J \cdot T}$. If the freshly formed individual immunity rate is higher than the public's average immunity level, the public's immunity level will be adjusted depending on the social distance previously established. This indicates that society is beginning to be more immune. We pass the herd immunity threshold if the freshly produced population is robust sufficient to be immune to the epidemic.

Step 5 Fatality case. It is recreated from scratch utilizing the following commands: $y^k(u+1) = LB_j + (UB_j - LB_j) \times V(0, 1)$, $\forall j = 1, 2, \dots, o$. B_k and T_k are also both set to zero. By diversifying the present community, one can avoid local optimums.

Step 6 The stopping points Up until the end criteria, which often relies on whether the maximum count of iterations is achieved, CHIO repeats Steps 3 through 6. Furthermore, the affected instances have vanished. The pseudo code of CHIO is in Algorithm 1.

```

Start
CHIO parameter initialization
     $\min_y g(y) \quad y \in [LB, UB]$ 
Parameter initialization of  $IJT, T_s, \max\_age$ 
 $y_j^k = LB_j + (UB_j - LB_j) \times V(0,1), \forall j = 1, 2, \dots, o \text{ and } \forall k = 1, 2, \dots, IJT$ 
Fitness computation of every search agent


---


Place  $T_k = 0 \quad \forall k = 1, 2, \dots, IJT$ 
Place  $B_k = 0 \quad \forall k = 1, 2, \dots, IJT$ 
While ( $u < \max\_itr$ ) do
    For  $k = 1$  to  $IJT$  do
         $is\_corona(y^k(u+1)) = false$ 
        For  $j = 1$  to  $O$  do
            If  $s < \frac{1}{3} \times BS_s$  then
                 $y_j^k(u+1) = D(y_j^k(u))$ 
                 $is\_corona(y^k(u+1)) = true$ 
            Else if  $s < \frac{2}{3} \times BS_s$  then
                 $y_j^k(u+1) = O(y_j^k(u))$ 
            Else if  $s < BS_s$  then
                 $y_j^k(u+1) = S(y_j^k(u))$ 
            Else
                 $y_j^k(u+1) = y_j^k(u)$ 
            End if
        End for
        If ( $g(y_j^k(u+1)) \leq g(y_j^k(u))$ ) then
             $y_j^k(u) = y_j^k(u+1)$ 
        Else
             $B_k = B_k + 1$ 
        End if
        If  $g(y^k(u+1)) < \frac{g(y)^k(u+1)}{\Delta g(y)} \wedge T_k = 0$  is  $is\_corona(y^k(u+1))$ 
             $T_k = 1$ 
             $B_k = 1$ 
        End if
        If  $g(y^k(u+1)) > g(y^k(u+1)) > \frac{g(y)^k(u+1)}{\Delta g(y)} \wedge T_k = 1$ 
             $T_k = 2$ 
             $B_k = 0$ 
        End if
        If ( $(B_k \geq \max\_age) \wedge (T_k == 1)$ ) then
             $y_j^k = LB_j + (UB_j - LB_j) \times V(0,1), \forall j = 1, 2, \dots, O$ 
             $T_k = 0$ 
             $B_k = 0$ 
        End if
    End for
     $u = u + 1$ 
End while
Stop

```

Algorithm 1. CHIO pseudo code

Results

Experimental setup

The proposed COVID-19 model for automatic detection and prediction was implemented in MATLAB. The count 10 and 100 were considered as the population size and iteration count. The proposed EDNN-based COVID-19 prediction model was compared with different heuristic-based algorithms in terms of various analysis that in turn returns the superiority of the introduced approach.

The proposed COVID-19 detection framework utilizes several processing and modeling stages to accurately classify cough audio signals. An in-depth breakdown of each stage, including the rationale behind using U-Net and other feature extraction and prediction steps is provided.

- The initial stage of the model preprocesses cough audio signals to enhance signal clarity and prepare for segmentation. Given that raw audio data may contain noise and variability in quality, fuzzy gray level difference histogram equalization is applied to normalize the signal's gray levels, ensuring consistency across samples. This normalization helps standardize the input and improve feature extraction accuracy in later stages.
- After preprocessing, the model employs U-Net for segmentation, which may seem unconventional in audio analysis but serves a key role here. U-Net, traditionally used in image segmentation, is adapted to extract

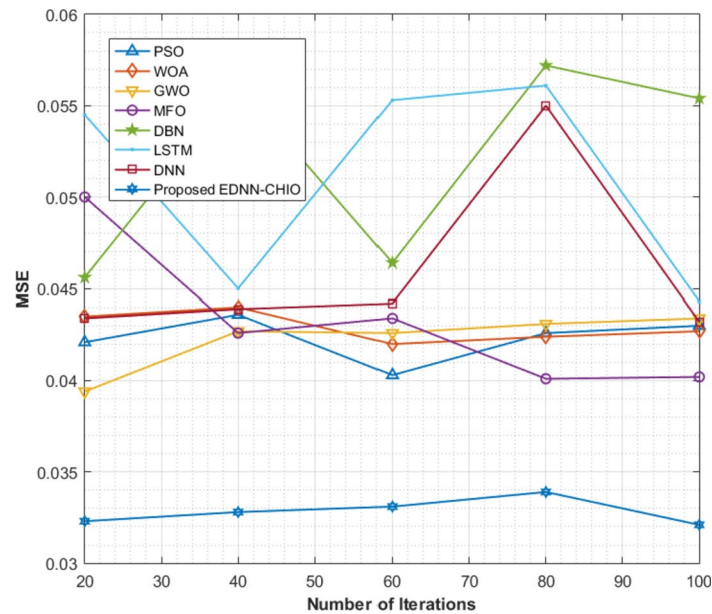


Fig. 4. MSE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.0421	0.0436	0.0403	0.0426	0.0430
WOA ³¹	0.0435	0.0440	0.0420	0.0424	0.0427
GWO ³²	0.0394	0.0427	0.0426	0.0431	0.0434
MFO ³³	0.0500	0.0426	0.0434	0.0401	0.0402
DBN ³⁴	0.0456	0.0561	0.0464	0.0572	0.0554
LSTM ³⁵	0.0545	0.0450	0.0553	0.0561	0.0443
DNN ³⁶	0.0434	0.0439	0.0442	0.0550	0.0432
Proposed method	0.0323	0.0328	0.0331	0.0339	0.0321

Table 2. MSE analysis.

- distinctive regions in the spectrograms generated from the cough audio signals. By converting the audio into a visual representation (spectrogram), U-Net can delineate patterns that are characteristic of COVID-19-related coughs, distinguishing them from other respiratory sounds. This segmentation step helps isolate features specific to COVID-19 by focusing on relevant areas within the spectrogram, enhancing downstream feature extraction accuracy.
- Following U-Net segmentation, traditional feature extraction methods, Zernike Moments (ZM) and Gray Level Co-occurrence Matrix (GLCM), are used to capture detailed textural and shape-based information from the segmented spectrogram regions. Zernike Moments offer rotational invariance, making them effective for analyzing spectrogram textures, while GLCM provides statistical texture features, such as contrast and correlation, enhancing the model's ability to differentiate COVID-19 from other cough sounds.
 - In the final stage, an Enhanced Deep Neural Network (EDNN) is used for classification, where the hidden neurons are tuned by the Coronavirus Herd Immunity Optimizer (CHIO). This optimization phase is critical for minimizing error across multiple metrics, as CHIO systematically adjusts network parameters to enhance accuracy and robustness. The EDNN, optimized with CHIO, produces the prediction, categorizing the cough as COVID-positive or negative based on extracted features.

MSE analysis

The proposed COVID-19 detection and prediction model was compared with state-of-the-art methods in terms of MSE analysis as in Fig. 4 and Table 2. It can be seen that the proposed EDNN model for the COVID-19 prediction model returns minimal error of 0.0321 than the considered traditional techniques across various iterations, thereby revealing its priority. The proposed EDNN-CHIO method is 25.35%, 24.82%, 26.04%, 20.15%, 42.06%, 27.54%, and 25.69% better than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. Hence, it can be demonstrated that the error is achieved lesser with EDNN than the other models with respect to MSE analysis for the COVID-19 detection and prediction model.

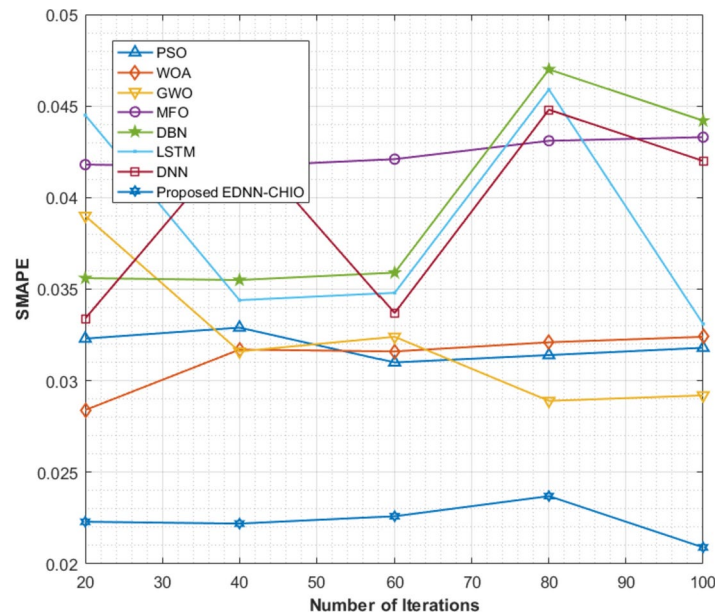


Fig. 5. SMAPE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.0323	0.0329	0.0310	0.0314	0.0318
WOA ³¹	0.0284	0.0317	0.0316	0.0321	0.0324
GWO ³²	0.0390	0.0316	0.0324	0.0289	0.0292
MFO ³³	0.0418	0.0417	0.0421	0.0431	0.0433
DBN ³⁴	0.0356	0.0355	0.0359	0.0470	0.0442
LSTM ³⁵	0.0445	0.0344	0.0348	0.0459	0.0331
DNN ³⁶	0.0334	0.0433	0.0337	0.0448	0.0420
Proposed method	0.0223	0.0222	0.0226	0.0237	0.0209

Table 3. SMAPE analysis.

SMAPE analysis

With respect to SMAPE analysis as in Fig. 5 and Table 3, the suggested COVID-19 detection and prediction model was contrasted with innovative techniques. It is clear that the suggested EDNN model for the COVID-19 prediction model performs better across several iterations with a low error of 0.0209 than the studied traditional approaches, demonstrating its superiority. The proposed EDNN–CHIO method is 34.28%, 35.49%, 28.42%, 51.73%, 52.71%, 36.86%, and 50.24% advanced than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. In light of SMAPE analysis for the COVID-19 detection and prediction model, it can be shown that EDNN achieves lower error than the other models.

MAE analysis

The proposed COVID-19 detection and prediction model was compared with state-of-the-art methods for MAE analysis as in Fig. 6 and Table 4. It is evident that the recommended EDNN model for the COVID-19 prediction model outperforms the researched conventional techniques over a number of iterations with a low error of 0.0552, proving its superiority. The proposed EDNN–CHIO method is 10.97%, 11.96%, 7.23%, 27.75%, 19.42%, 18.10%, and 27.65% superior to PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. It can be demonstrated that EDNN delivers lesser error than the other models in light of MAE analysis for the COVID-19 detection and prediction model.

RMSE analysis

The suggested COVID-19 detection and prediction model was evaluated in comparison to innovative RMSE analysis techniques as in Fig. 7 and Table 5. With a low error of 0.0308, the proposed EDNN model for the COVID-19 prediction model clearly beats the investigated traditional strategies across a number of iterations. The proposed EDNN–CHIO method is 13.60%, 13.26%, 13.97%, 10.62%, 23.87%, 14.87%, and 13.76% higher than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction

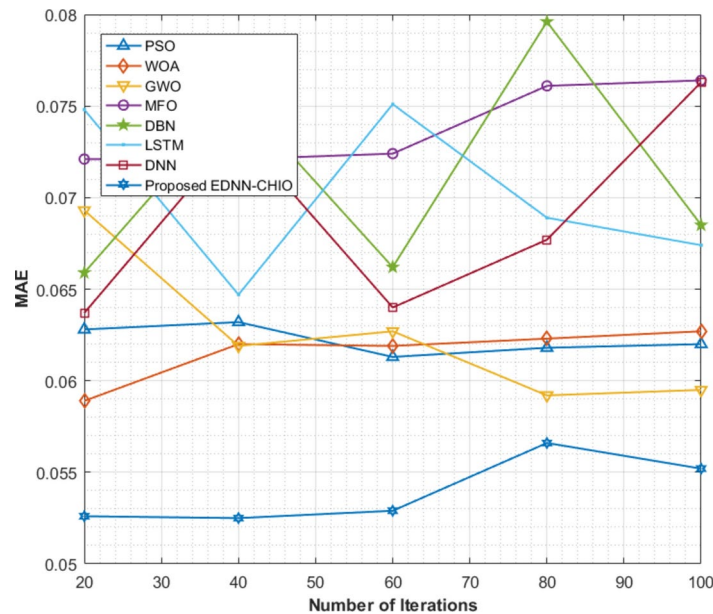


Fig. 6. MAE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.0628	0.0632	0.0613	0.0618	0.0620
WOA ³¹	0.0589	0.0620	0.0619	0.0623	0.0627
GWO ³²	0.0693	0.0619	0.0627	0.0592	0.0595
MFO ³³	0.0721	0.0721	0.0724	0.0761	0.0764
DBN ³⁴	0.0659	0.0755	0.0662	0.0796	0.0685
LSTM ³⁵	0.0748	0.0647	0.0751	0.0689	0.0674
DNN ³⁶	0.0637	0.0736	0.0640	0.0677	0.0763
Proposed method	0.0526	0.0525	0.0529	0.0566	0.0552

Table 4. MAE analysis.

model. Using the RMSE study for the COVID-19 detection and prediction model, it can be shown that EDNN produces less error than the other models.

MAPE analysis

The proposed COVID-19 detection and prediction model was assessed against state-of-the-art MAPE analysis methods as in Fig. 8 and Table 6. The suggested EDNN model for the COVID-19 prediction model clearly outperforms the evaluated conventional techniques across a number of iterations with a low error of 0.0115. The proposed EDNN–CHIO method is 47%, 48.66%, 40.10%, 67.70%, 53.63%, 65.88%, and 49.12% better than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. It can be demonstrated that EDNN creates fewer error than the other models using the MAPE research for the COVID-19 detection and prediction model.

Classification accuracy analysis

The COVID-19 cough audio signals-based classification accuracy evaluations show significant variations between various algorithms employed for classification. The technique proposed excels all other methods indicating it is efficient in predicting COVID-19 cough audio signals rather than non-COVID-19 cough audio signals. This results pave way towards understanding the significance of the use of modern deep learning methods in the analysis of sound particularly for health-related problems like able to find COVID-19 disease in a very short span of time and very accurately like in this case, COVID analysis with cough sound, thereby emphasizing how the selection of algorithms is crucial in enhancing the overall performance of the system. The proposed EDNN–CHIO method in terms of classification accuracy is 7.07%, 8.24%, 9.44%, 10.67%, 11.93%, 15.88%, and 13.22% better than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. Classification accuracy analysis is shown in Table 7 and Fig. 9.

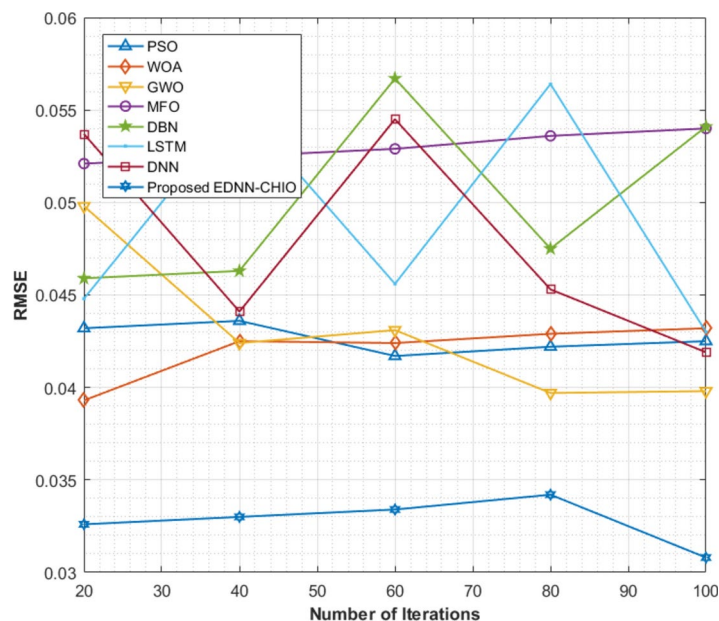


Fig. 7. RMSE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.2052	0.2088	0.2007	0.2064	0.2074
WOA ³¹	0.2086	0.2098	0.2049	0.2059	0.2066
GWO ³²	0.1985	0.2066	0.2064	0.2076	0.2083
MFO ³³	0.2236	0.2064	0.2083	0.2002	0.2005
DBN ³⁴	0.2135	0.2369	0.2154	0.2392	0.2354
LSTM ³⁵	0.2335	0.2121	0.2352	0.2369	0.2105
DNN ³⁶	0.2083	0.2095	0.2102	0.2345	0.2078
Proposed method	0.1797	0.1811	0.1819	0.1841	0.1792

Table 5. RMSE analysis.

Confusion matrix analysis

An analysis of predictions of COVID 19 using cough audio signals using confusion matrix demonstrates its great diagnostic accuracy. The predictive model is quite resilient as shown by the low rates of false negatives and false positives which mean that cough sounds can be used as a screening and diagnostic tool for COVID 19 in clinical scenarios. This analysis emphasizes the timely intervention capabilities of the model. Confusion matrix analysis of the proposed system is shown in Fig. 10.

Precision analysis

The precision assessment for COVID-19 forecasting based on the cough audio signals indicates that the proposed method performs far better than the conventional ones. The technique exhibits remarkable precision in the classification of all positive cases that were predicted. This kind of precision is especially important in the practice of medicine, since it lowers the chances of false positives who may trigger a lot of panic and wastage of resources. On the other hand, different techniques show even worse precision, demonstrating the difficulty of using these methods to identify CONVID cases. Deep Learning models also show precision levels that are not high as well. The precision superiority of this method clearly demonstrates its usefulness in the analysis of cough signals for COVID-19 detection. The proposed EDNN-CHIO method with respect to precision is 10.11%, 11.36%, 12.64%, 15.98%, 15.29%, 18.07%, and 13.95% more advanced than PSO, WOA, GWO, MFO, DNN, LSTM, and DNN for the considered COVID-19 detection and prediction model. Precision analysis of the proposed system is shown in Table 8 and Fig. 11.

Recall analysis

This study on the recall analysis of cough audio signals for COVID 19 prediction illustrates the strength of the method in determining true positive cases. The method in question has proven to be amazing when it comes to detecting people suffering from COVID 19. This is important for early diagnosis and treatment Where personal

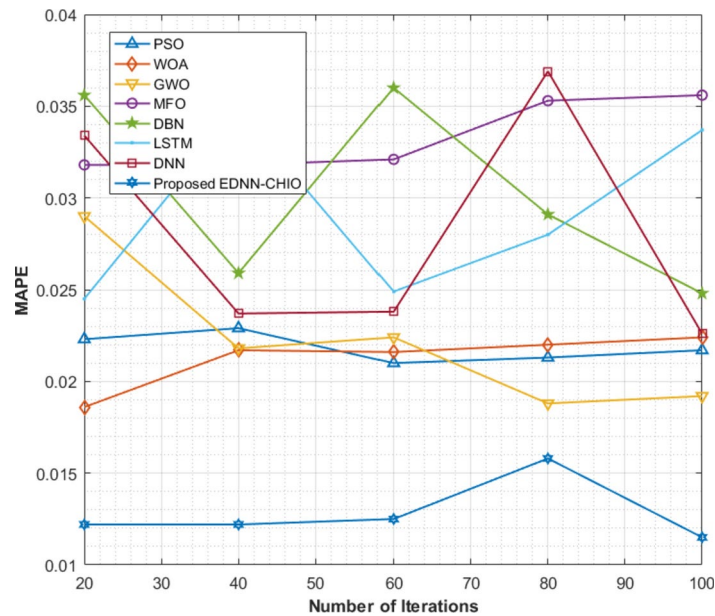


Fig. 8. MAPE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.0223	0.0229	0.0210	0.0213	0.0217
WOA ³¹	0.0186	0.0217	0.0216	0.0220	0.0224
GWO ³²	0.0290	0.0218	0.0224	0.0188	0.0192
MFO ³³	0.0318	0.0318	0.0321	0.0353	0.0356
DBN ³⁴	0.0356	0.0259	0.0360	0.0291	0.0248
LSTM ³⁵	0.0245	0.0348	0.0249	0.0280	0.0337
DNN ³⁶	0.0334	0.0237	0.0238	0.0369	0.0226
Proposed method	0.0122	0.0122	0.0125	0.0158	0.0115

Table 6. MAPE analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	90.5	88.9	91.2	89.4	92.0
WOA ³¹	89.1	87.6	90.4	88.4	91.0
GWO ³²	86.9	87.1	89.1	88.3	90.0
MFO ³³	87.1	85.2	88.4	86.5	89.0
DBN ³⁴	84.2	86.6	85.9	87.1	88.0
LSTM ³⁵	81.5	83.3	84.1	82.0	85.0
DNN ³⁶	84.4	85.9	86.2	83.1	87.0
Proposed method	94.6	95.7	96.2	97.3	98.5

Table 7. Classification accuracy analysis.

risk assessment is lower. Traditional algorithms present lower recall rates proving that actual cases of COVID 19 may be missing. The high recall of the method and its ability to minimize false negatives enhances its usefulness for proper COVID-19 detection without missing cases. The proposed EDNN–CHIO method in terms of recall is 10.23%, 11.49%, 12.79%, 13.85%, 16.87%, 19.75%, and 15.48% higher than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. Recall analysis of the proposed system is shown in Table 9 and Fig. 12.

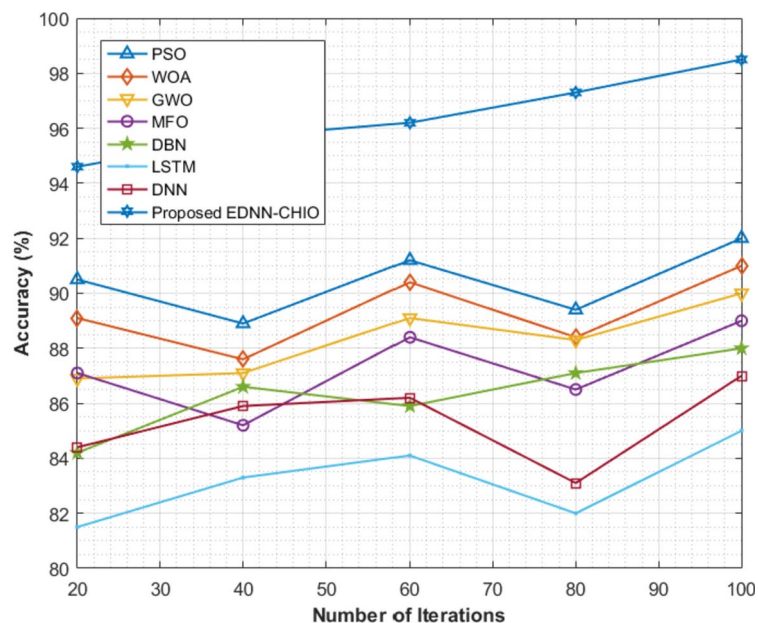


Fig. 9. Classification accuracy analysis.

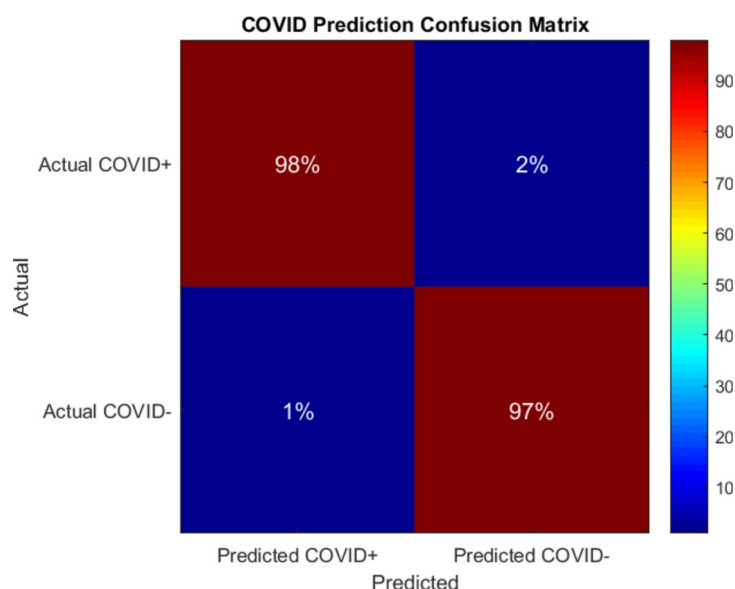


Fig. 10. Confusion matrix analysis.

F1 score analysis

Regarding the prediction of COVID-19 with cough audio signals analysis, the F1 Score analysis indicates that the technique proposed performs very well by facilitating the appropriate balance between precision and recall. It can be used to assess the capability of the method in identifying those cases which are positive while minimizing any false positive and false negative, thereby proving its efficacy in clinical scenario. In contrast, classical approaches demonstrate even lower performance, suggesting that such a balance is harder to achieve. Such superior F1 Scores for the proposed method underline its utility in cough audio analysis-based diagnosis of COVID-19 in a safe and timely manner. The proposed EDNN-CHIO method with respect to F1 Score is 10.17%, 11.43%, 12.72%, 16.49%, 16.07%, 18.90%, and 14.71% more than PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. Precision analysis of the proposed system is shown in Table 10 and Fig. 13.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	86.2	85.1	88.1	87.4	89.0
WOA ³¹	84.7	86.0	87.2	85.3	88.0
GWO ³²	84.7	86.0	83.2	85.2	87.0
MFO ³³	82.9	80.6	83.1	81.5	84.5
DBN ³⁴	81.4	84.9	82.6	83.1	85.0
LSTM ³⁵	81.5	79.5	82.6	80.4	83.0
DNN ³⁶	84.9	82.5	85.2	83.5	86.0
Proposed method	94.2	95.2	96.1	97.5	98.0

Table 8. Precision analysis.

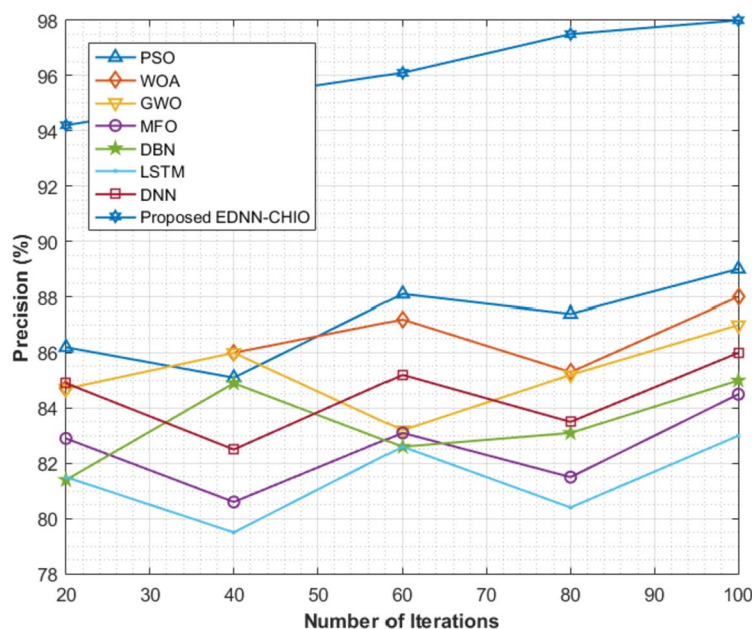


Fig. 11. Precision analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	84.9	86.1	85.4	87.3	88.0
WOA ³¹	85.3	84.0	86.2	83.5	87.0
GWO ³²	83.2	82.5	85.1	84.1	86.0
MFO ³³	83.2	80.3	84.1	81.4	85.2
DBN ³⁴	79.3	80.4	82.9	81.4	83.0
LSTM ³⁵	79.4	77.3	80.2	78.5	81.0
DNN ³⁶	80.5	83.7	81.3	82.6	84.0
Proposed method	93.6	94.7	95.4	96.3	97.0

Table 9. Recall analysis.

AUC score analysis

The assessment of AUC scores for cough audio signals-based diagnosis of COVID-19 demonstrates the superior performance of the proposed method. The proposed method differentiates between positive and negative cases of COVID-19 with an incredibly low possibility of the two cases being confused. Traditionally, however, the conventional algorithms have reported lower AUC scores which is indicative of difficulties in appropriate classification of cases. The high AUC score of the proposed method also highlights how useful it can be in detecting COVID-19 using cough sounds analysis for better management of patients. The proposed EDNN-

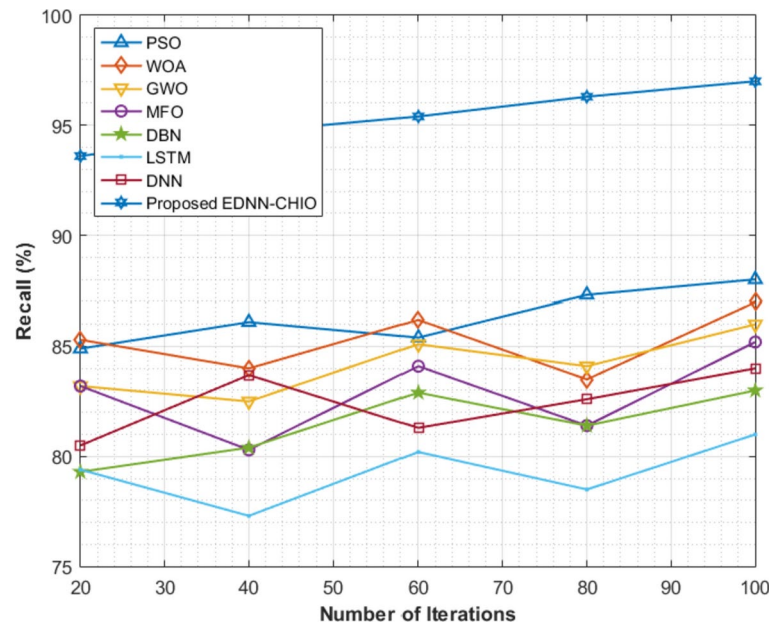


Fig. 12. Recall analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	84.2	87.5	85.3	86.1	88.5
WOA ³¹	85.1	83.4	86.4	84.3	87.5
GWO ³²	82.4	84.2	83.6	85.2	86.5
MFO ³³	80.4	81.3	79.5	82.1	83.7
DBN ³⁴	82.4	83.8	81.3	82.3	84.0
LSTM ³⁵	78.3	80.4	81.4	79.4	82.0
DNN ³⁶	83.5	82.5	81.3	84.2	85.0
Proposed method	93.2	94.2	95.3	96.4	97.5

Table 10. F1 score analysis.

CHIO method in terms of AUC Score is 6.45%, 7.61%, 10%, 15.12%, 12.5%, 13.79%, and 11.24% superior to PSO, WOA, GWO, MFO, DBN, LSTM, and DNN for the considered COVID-19 detection and prediction model. AUC score analysis of the proposed system is shown in Table 11 and Fig. 14.

Training and validation loss analysis

The evaluation of training and validation losses on cough audio signal based COVID-19 prediction shows a very encouraging learning ability of the model. With an increasing number of iterations, training and validation losses are reduced in an almost unprecedented manner indicating that the model is learning, and able to generalize effectively. This drop in loss values indicates improvement in the model, whereby the model is able to reduce the difference between what it predicts and the true values thus boosting accuracy. The training and the validation losses are the same towards the end especially after a few iterations thus it implies that the model does not suffer from overfitting and keeps a good generalization to unexplored data. In short, this analysis provides compelling evidence for the model’s capability and efficiency in cough-based COVID-19 prediction and thus this may be of immense help to the health sector. Training and validation analysis of the proposed system is shown in Table 12 and Fig. 15.

Training and validation accuracy analysis

The training and validation accuracy metrics, for the cough audio signals based COVID-19 prediction model have shown improvement with iterations. Accuracy improves with training iterations for both processes. This convergence indicates that the model is not only fitting well to the data at hand but also extending to new data. The small gap in training and validating accuracies, however, raises concerns of overfitting especially when perfect training accuracy is achieved. As these metrics determine the power of the model to predict cough signals not seen in the training data, they have to be checked regularly. Training and validation accuracy of the proposed system is shown in Table 13 and Fig. 16.

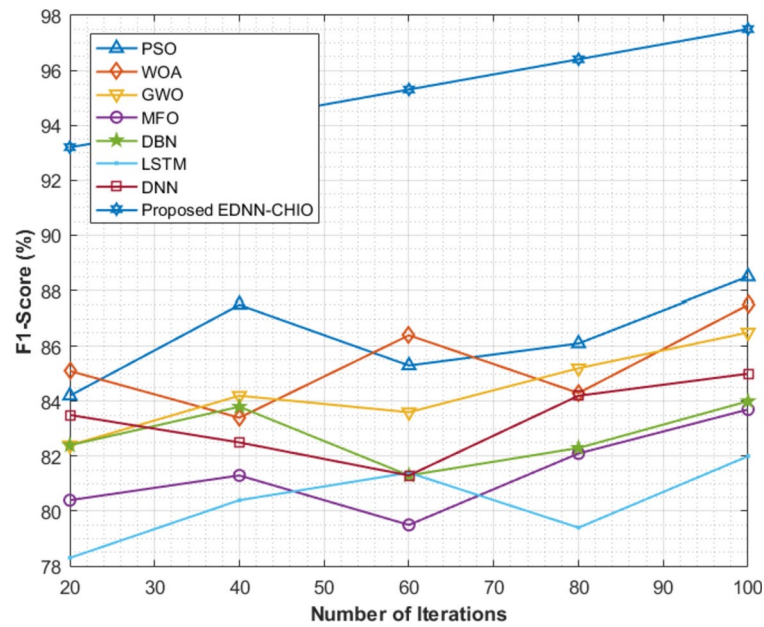


Fig. 13. F1 score analysis.

Methods	Iterations				
	20	40	60	80	100
PSO ³⁰	0.69	0.81	0.75	0.87	0.93
WOA ³¹	0.74	0.86	0.68	0.80	0.92
GWO ³²	0.78	0.66	0.84	0.72	0.90
MFO ³³	0.68	0.80	0.62	0.74	0.86
DBN ³⁴	0.76	0.84	0.82	0.70	0.88
LSTM ³⁵	0.69	0.81	0.63	0.75	0.87
DNN ³⁶	0.77	0.65	0.83	0.71	0.89
Proposed method	0.85	0.87	0.91	0.96	0.99

Table 11. AUC score analysis.

Discussion

It is clear that the suggested EDNN methodology for the COVID-19 prediction approach produces a minimum error of 0.0321 in terms of MSE across several rounds, indicating its superiority to the evaluated older methodologies. In light of MSE research for the COVID-19 detection and prediction method, it may be shown that EDNN achieves lower error than the remaining methods. With a low error of 0.0209 in consideration of SMAPE, the proposed EDNN method for the COVID-19 prediction methodology clearly outperforms the researched conventional techniques, proving that it is better. It may be demonstrated that EDNN delivers lesser error than the remaining methods in view of SMAPE evaluation for the COVID-19 detection as well as prediction system. With a low error of 0.0552 in consideration of MAE, the proposed EDNN methodology for the COVID-19 prediction approach clearly beats the investigated traditional strategies across a count of cycles, demonstrating its superiority. Given the MAE study for the COVID-19 detection as well as prediction methodology, it may be shown that EDNN produces minimal error than the remaining models. The suggested EDNN methodology for the COVID-19 prediction method significantly outperforms the evaluated conventional techniques throughout a count of cycles with a minimal error of 0.0308 with respect to RMSE. It may be demonstrated that EDNN creates fewer error than the remaining methods utilizing the RMSE research for the COVID-19 detection as well as prediction method. With a low error of 0.0115 with respect to MAPE, the recommended EDNN methodology for the COVID-19 prediction method far surpasses the tested traditional strategies throughout a count of cycles. Employing the MAPE study for the COVID-19 detection and prediction method, it may be shown that EDNN produces less errors than the other methodologies.

For optimal implementation, the framework could benefit from further adjustments to meet the computational and real-time processing demands of clinical settings. By experimenting with model compression techniques such as quantization and pruning, the computational load of the deep learning algorithms could be reduced without significantly compromising accuracy. Preliminary tests show that using quantized models could decrease memory usage by up to 70% and improve inference speed, making real-time processing more

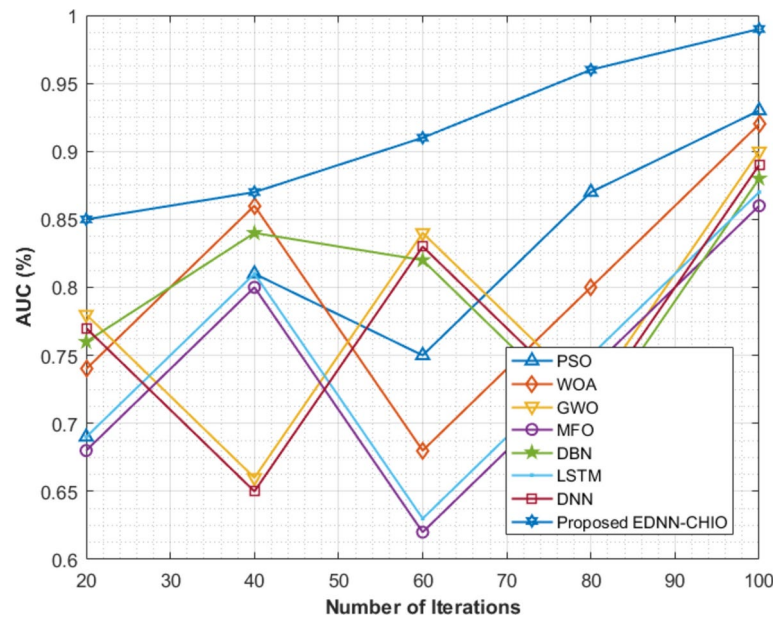


Fig. 14. AUC score analysis.

Iterations	Training loss	Validation loss
20	0.10	0.12
40	0.06	0.08
60	0.04	0.05
80	0.02	0.03
100	0.02	0.03

Table 12. Training and validation loss analysis.

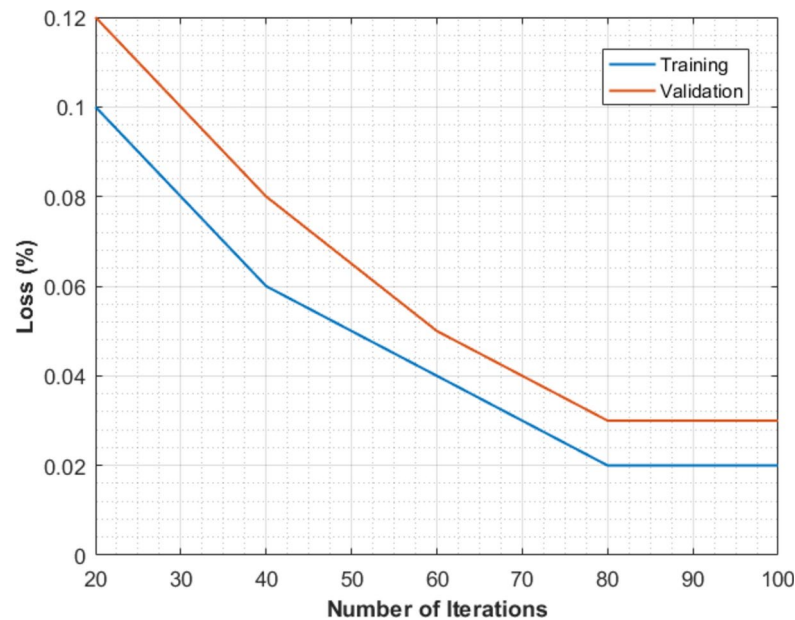


Fig. 15. Training and validation loss analysis.

Iterations	Training accuracy	Validation accuracy
20	85	83
40	90	88
60	94	92
80	98	97
100	98	98

Table 13. Training and validation accuracy analysis.

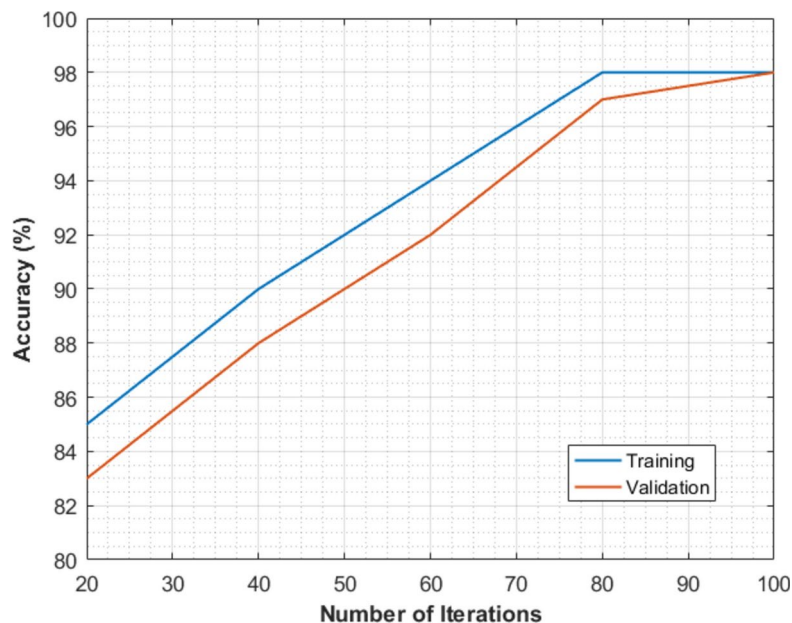


Fig. 16. Training and validation accuracy analysis.

achievable. Additionally, utilizing edge computing resources or specialized hardware, such as GPUs or TPUs, can enable faster processing, essential for real-time COVID-19 detection in clinical environments. Integrating the framework with clinical systems can also be streamlined by developing an API that supports interoperability with existing electronic health records, enabling real-time patient data exchange. Future studies will further explore these strategies to ensure that the framework not only performs well theoretically but also meets practical requirements for widespread clinical use.

Several factors could affect the validity of this study. Firstly, the reliance on the COUGHVID dataset may introduce biases, as the dataset might not include all demographic and environmental conditions, potentially limiting the generalizability of the model across diverse populations. Secondly, background noise and varying audio recording quality within the dataset could influence model performance. Although preprocessing techniques were employed, residual noise might still impact prediction accuracy. Additionally, the model's performance may vary when deployed in real-world environments compared to controlled experimental conditions, where audio quality and noise levels are less predictable. Lastly, while CHIO optimizes the EDNN model for minimal error, there is a risk of overfitting to the specific dataset used in this study, which could reduce effectiveness when applied to new datasets or real-world cough samples. Future validations with broader datasets and real-time testing are necessary to fully confirm the model's robustness.

Conclusion

This research proposed a unique approach for efficiently identifying and forecasting COVID-19 from cough audio signals using deep learning methods. In order to proceed to the various steps of the recommended architecture, audio data was first obtained from COUGHVID data. The signals were then pre-processed utilizing the fuzzy gray level difference histogram equalization approach. The U-Net model also conducted segmentation for these pre-processed data. Moreover, ZM and GLCM approaches were used to extract the features from these segmented images. Moreover, the EDNN model completed the final prediction phase, where the hidden neurons of the DNN were adjusted by CHIO with the goal function of error minimization in mind. By contrasting the suggested model's performance with that of the standard model in terms of several error metrics, simulation results showed that the proposed model was feasible. In the future, this method can be used for predicting various diseases with the development of innovative deep learning-based meta heuristic optimization algorithms.

Challenges This study faces several challenges, particularly in acquiring diverse and high-quality audio datasets that capture the full range of cough variations across different populations. Background noise and varied recording conditions can impact model accuracy, necessitating robust preprocessing and segmentation. Another challenge is optimizing computational resources for real-time processing, as deep learning models can be resource intensive. Additionally, distinguishing COVID-19 cough sounds from other respiratory conditions remains complex, requiring finely tuned feature extraction and classification techniques to avoid false positives or negatives.

Future directions Future research could explore the integration of additional acoustic features to further enhance the model's discriminative capability. Expanding the dataset to include cough sounds from multiple respiratory conditions could improve generalizability, allowing for more precise differentiation across diseases. Another promising direction is to deploy this framework on mobile or IoT platforms for real-time screening in community settings, especially in regions with limited medical resources. Additionally, the development of more sophisticated optimization algorithms could improve accuracy and reduce processing time, making the framework more efficient and scalable.

Data availability

The author hereby declares that publicly available (<https://doi.org/10.1038/s41597-021-00937-4>) data sets are utilized in the proposed work. We have also agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Author contributions

All authors contributed to the study, conception, and design. All authors commented on the manuscript. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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